

FILTER DESIGN TEST CASE

Coupled Resonator Filter - Band 12 (700 MHz) for Public Safety Band Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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Design Center:	Irvine Design Center
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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Coupled Resonator Filter filter targeting Band 12 (700 MHz) for Public Safety Band Filter. The filter is designed on Langasite substrate with a target center frequency of 4942.6 MHz and bandwidth of 123.4 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Public Safety Band Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Coupled Resonator Filter
- Frequency Band: Band 12 (700 MHz)
- Application: Public Safety Band Filter
- Substrate: Langasite
- Package: LGA (Land Grid Array)
- Design Tool: Matlab RF Toolbox R2024b
- Design Center: Irvine Design Center

This specification applies to all variants of the Coupled Resonator Filter design targeting Band 12 (700 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	4942.6 MHz
Bandwidth (3 dB):	123.4 MHz
Insertion Loss Target:	<= 2.8 dB
Return Loss Target:	>= 16.0 dB
Out-of-Band Rejection:	>= 50.0 dB
Isolation (Duplexer):	>= 35.0 dB
Q Factor:	677
Temperature Coefficient:	-16.2 ppm/°C
Die Size:	1.3 x 1.9 mm
Wafer Size:	8-inch
Number of Resonators:	7
Electrode Material:	Mo/Al

Passivation: SiO2

4. TEST PROCEDURE

1. Open Matlab RF Toolbox R2024b and load the Coupled Resonator Filter template project.
2. Define the substrate stack: Langasite with specified layer thicknesses.
3. Set design targets: center freq 4942.6 MHz, BW 123.4 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 2.8 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (LGA (Land Grid Array)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, Langasite).
10. Perform on-wafer probing using Anritsu MS46122B ShockLine.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 14828 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.8 dB
- Return loss in passband shall be >= 16.0 dB
- Out-of-band rejection at +/- 185 MHz offset >= 50.0 dB
- Group delay variation across passband shall be <= 4 ns
- Temperature drift over -40°C to +85°C shall be <= 10.0 MHz
- Power handling shall be >= +32 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 91%

6. TEST RESULTS

Measurement Date: 2025-06-01
Devices Tested: 15
Insertion Loss (meas): 2.55 dB
Return Loss (meas): 18.1 dB
Rejection (meas): 52.8 dB
Isolation (meas): 40.3 dB
Q Factor (meas): 677
Group Delay Variation: 1.2 ns
Power Handling: +36 dBm
Die Yield: 90.1%

Result Classification: CONDITIONAL PASS

7. OBSERVATIONS AND FINDINGS

- a) The Coupled Resonator Filter design demonstrated acceptable performance for Band 12 (700 MHz).
- b) Insertion loss is marginally above target; electrode thickness optimization recommended.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Slight frequency drift observed at +85°C; TC layer thickness may need adjustment.
- e) ESD robustness exceeded specification by 2x margin.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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